Intelligent Robotics - M.EIC041
2024/2025 - 1S

Assignement 4:

Towards LLM-Based Geometric Localization in a Simulated Environment

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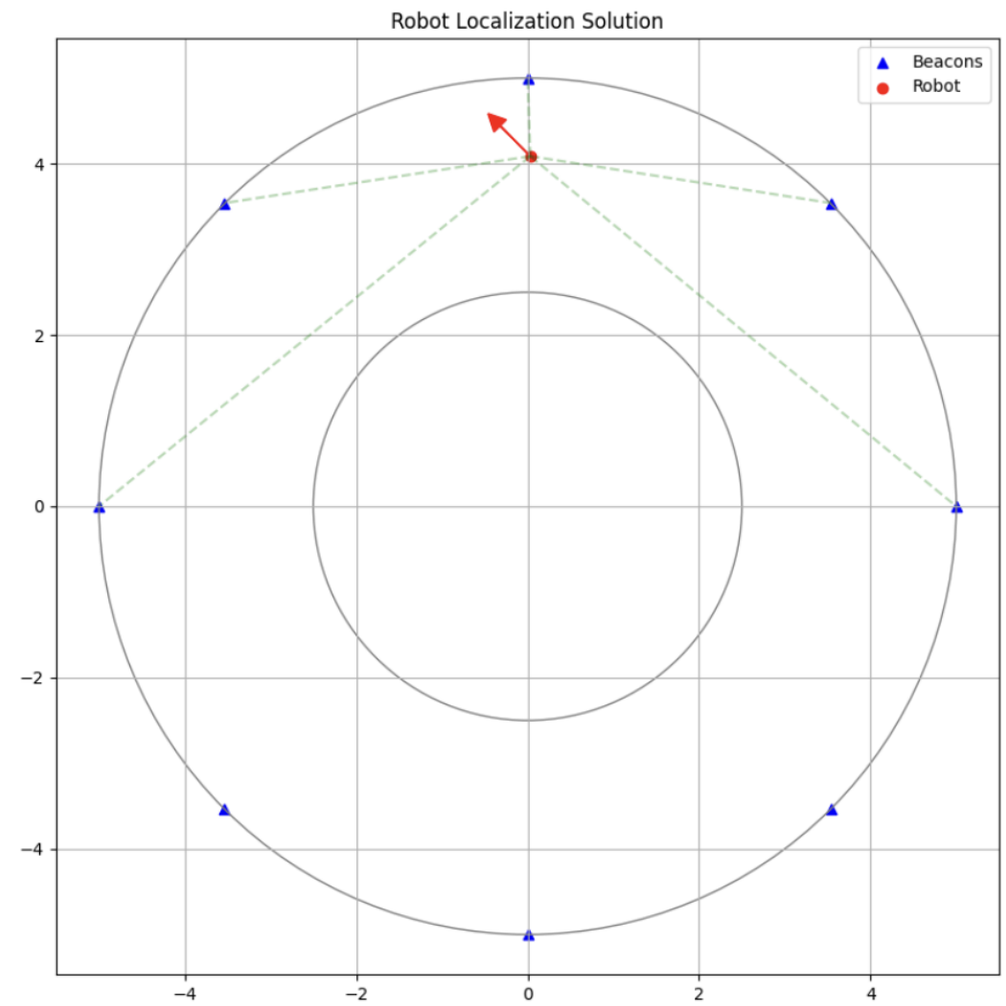
Introduction & Motivation

LLM-based robot localization

- Critical need for accurate positioning in Mobile Robotics
- LLMs show promise in robotics planning & understanding
- Gap: LLM effectiveness in pose estimation unknown
- Project goal: Test LLMs for robot localization



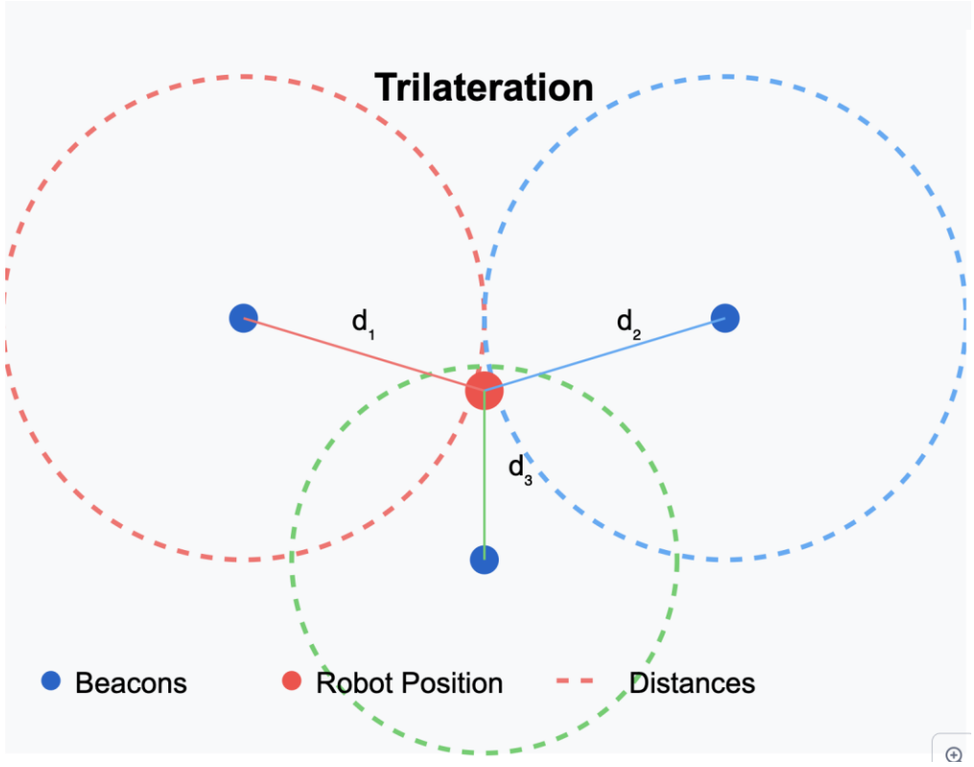
State of the Art

Mobile Robot Localization

- **Probabilistic Filters (Smith & Cheeseman, 1986):**
- Maintains Gaussian (EKF) or particle-based (Monte Carlo) belief distributions of robot state

- **Deep Learning SLAM (Wang et al., 2017):**
- Neural networks enhance geometric feature extraction from raw sensor data

- **Analytical Geometric Methods (Leonard & Durrant-Whyte, 1991):**
- Closed-form or optimization-based solutions using beacon measurements (n ≥ 3)

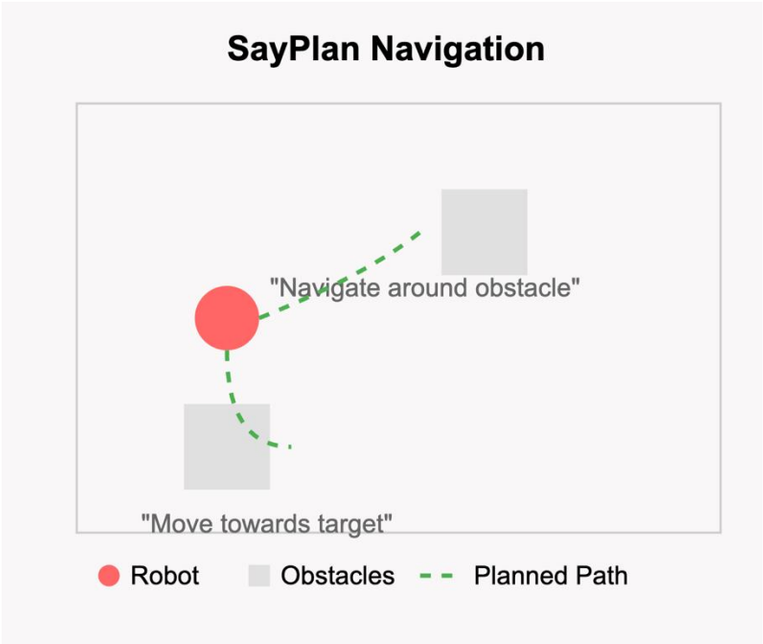


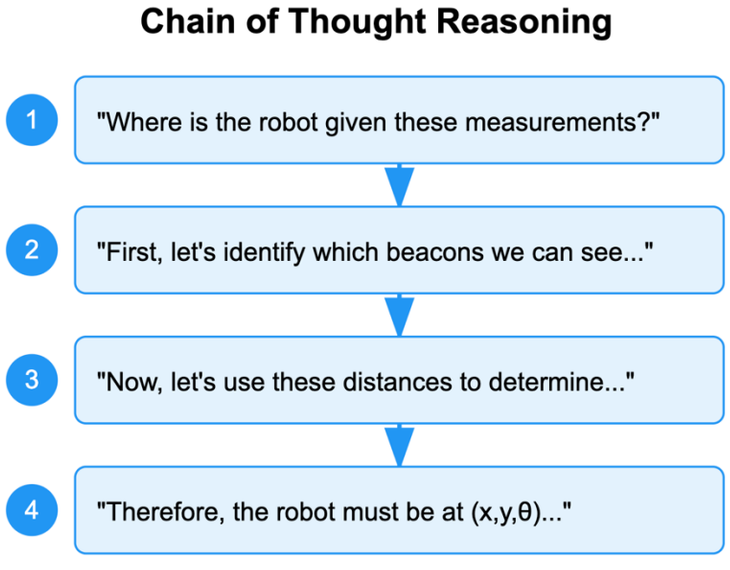
State of the Art

Large Language Models in Mobile Robotics

- **SayPlan (Rana et al., 2023):**
- Demonstrates LLMs can understand and generate navigation plans from spatial scene descriptions
- Shows LLMs can handle complex geometric constraints and relations

- **Chain-of-Thought Prompting (Wei et al., 2022):**
- Improves mathematical reasoning by breaking problems into explicit steps
- Enables validation and error checking through step-by-step deduction





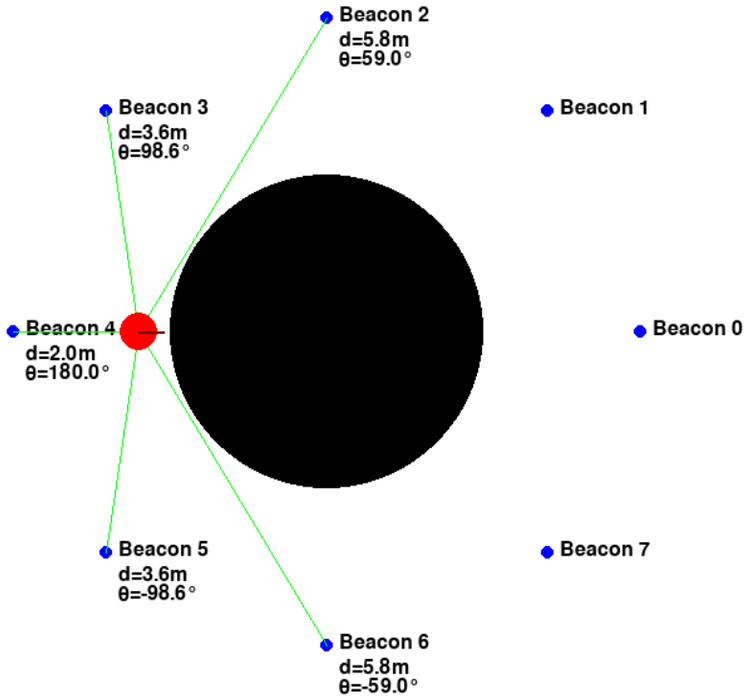
Proposed Framework

Towards LLM-Based Geometric Localization in a Simulated Environment

- **2D Arena Design:**
- Circular wall (r=2.5m) creates visibility constraints in a structured environment 8 fixed beacons arranged in outer circle (r=5.0m) provide reference points

- **Robot Sensor Model:**
- Measures distances and angles to visible beacons in robot frame
- Wall occlusion creates realistic beacon visibility constraints

- **LLM Integration:**
- Geometric reasoning through structured prompts
- Step-by-step triangulation and validation of pose estimates Cross-validation using multiple beacon measurements



Experimental Evaluation and Design

Experimental Setup

4 state-of-the-art language models:

- Claude 3.5 Sonnet (October 2024)
- Gemini Pro 2.0
- GPT-4 Turbo (January 2024)
- Llama 2 70B

Test Scenarios

Six distinct localization scenarios with different complexity:

- Ex1) Basic case with symmetric beacon visibility $(x = -3.0,\ y = 0.0,\ \theta = 0.0^\circ)$
- Ex2) Edge case near beacon radius $(x = -3.4,\ y = 0.0,\ \theta = 90.0^\circ)$
- Ex3) Constrained visibility $(x = -0.05,\ y = -3.75,\ \theta = 90.0^\circ)$
- Ex4) Complex orientation $(x = 2.96,\ y = -2.33,\ \theta = 135.0^\circ)$
- Ex5) Limited beacon visibility $(x = 2.04,\ y = 1.88,\ \theta = 90.0^\circ)$
- Ex6) Maximum distance case $(x = -4.39,\ y = -1.89,\ \theta = 175.9^\circ)$

Experimental Evaluation and Design

Prompting Strategies

Each model performed 2D robot localization using **three** distinct prompting strategies.

- **Standard**: Technical environment and task description
- **Math**: Optimization-based formulation with formal constraints
- **Reasoning**: Intuitive geometric problem-solving framework

Standard Prompt Development

- Focus on pure technical specification without guiding the solution approach

Definition of the complete state space through the robot's configuration vector (x, y, θ).

Two critical geometric constraints:

- A central wall creating visibility occlusions
- An operating radius limiting valid positions

Presence of eight equidistant beacons

Inclusion of a solved example

Experimental Evaluation and Design

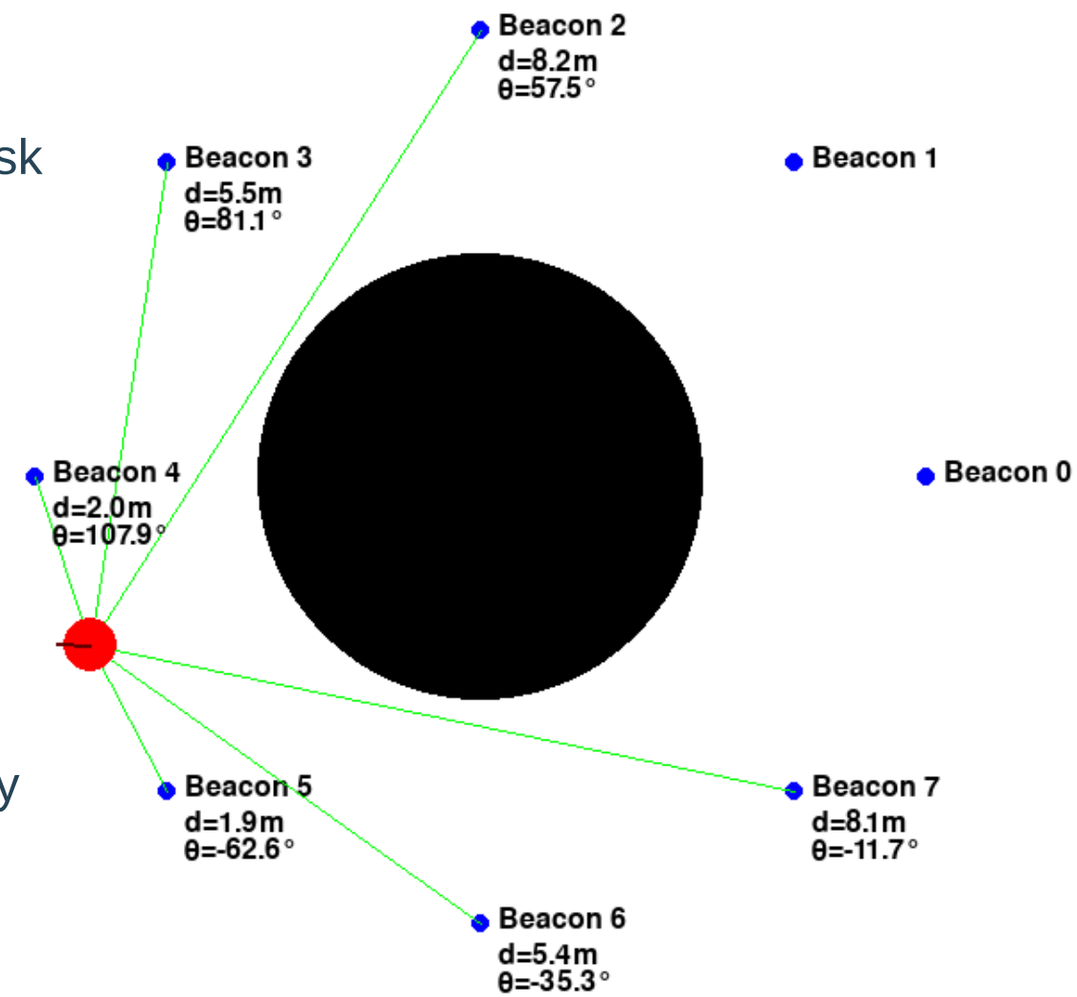
Mathematical Prompt Adaptation

Formulating the localization problem as an optimization task complete state estimation including orientation

- Guides LLMs toward analytical solution using a mathematical formulation of the problem
- Minimizes the error in predicted and observed measurements

Incorporates

- Full state vector estimation
- Explicit geometric constraints affecting beacon visibility



Experimental Evaluation and Design

Reasoning Prompt Adaptation

Use of spatial relationships and systematic deduction

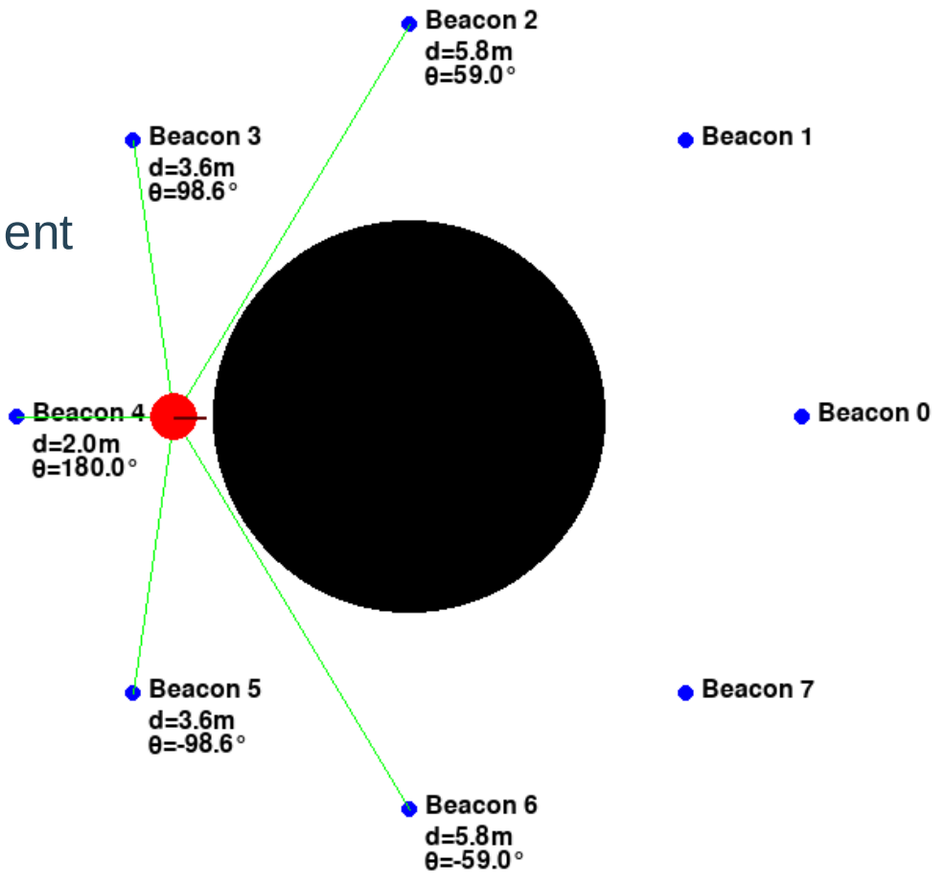
Emphasis on symmetry, perspectivity,

- Beacon configurations described through relative positions
- Emphasis on the qualitative characteristics of the environment
- Use of reasoning based on perspectives, logic symmetries

Visibility patterns as geometric clues

Incorporates

- Full state vector estimation
- Explicit geometric constraints affecting beacon visibility



Experimental Evaluation and Design

Evaluation Metrics

How performance was evaluated:

- Euclidean position error
- Absolute angular error
- Percentage of visible beacons correctly processed

Experimental Evaluation and Design

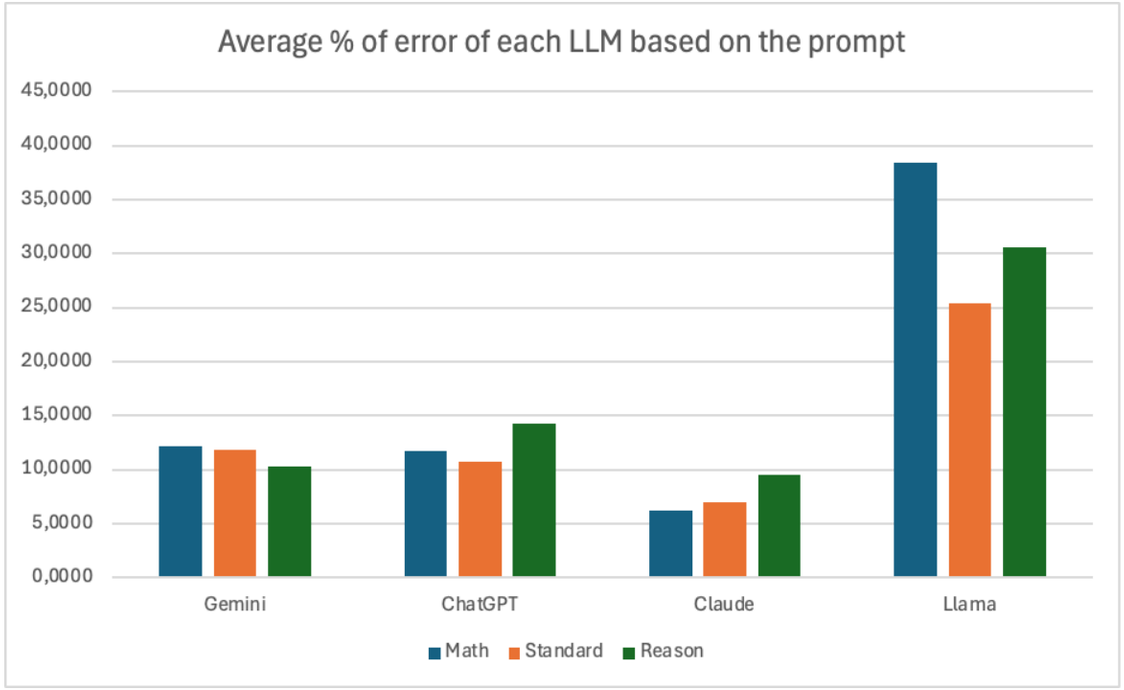
Result of the experiment

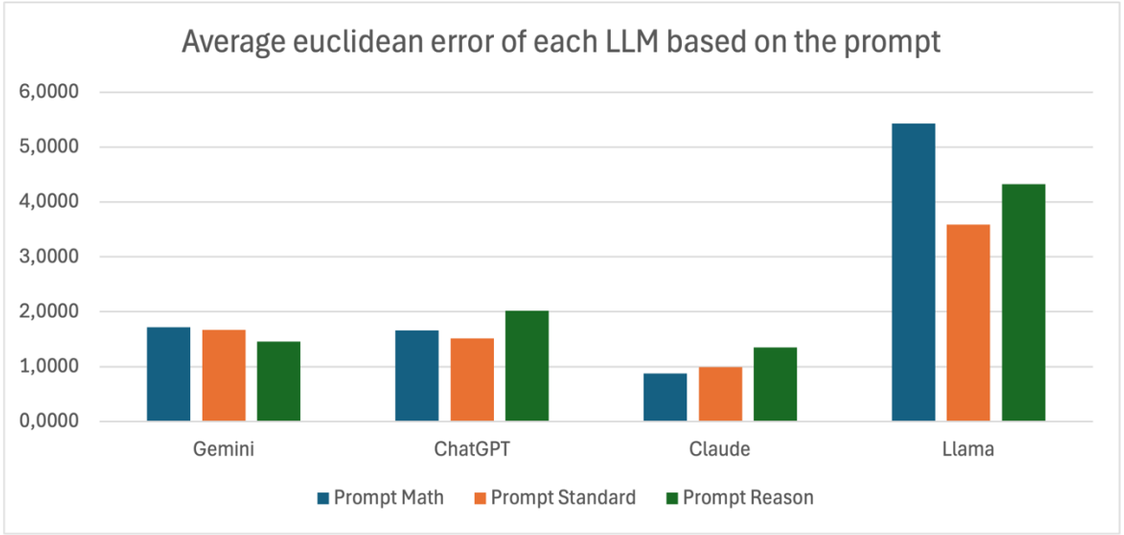
Accuracy

- **The LLM :**
- Mainly due to their type, performance varies heavily

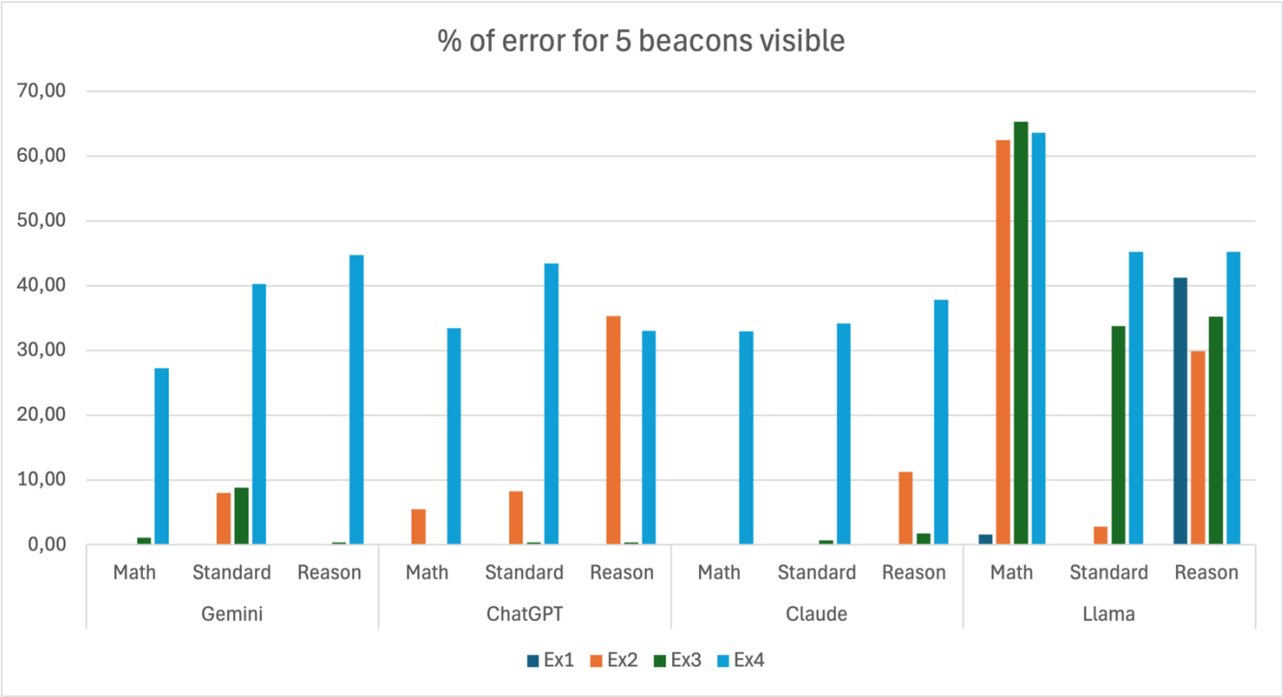
- **The prompt:**
- It appears that some LLM or better with math than others, but our original prompt is always between math and reason

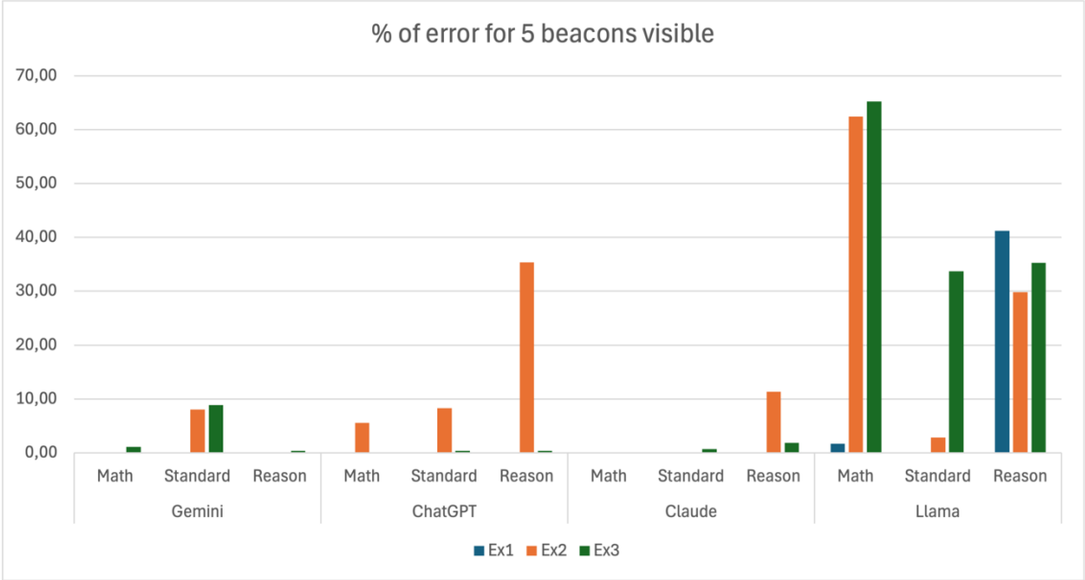
- **What we can understand of it :**
- Let's take a deep look into the data

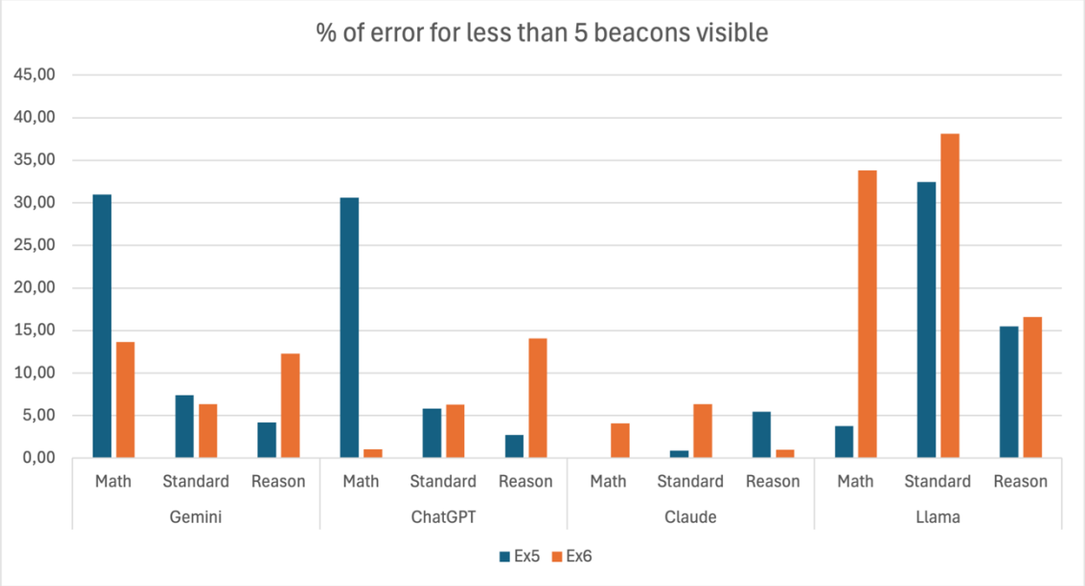




Result of the experiment







Future works

The main ideas

- **Add noises :**
- To be able to see the real power of each LLM

- **Mix prompts:**
- To see if by mixing some part of each prompt we can improve the results of everything

- **Make it work on an actual robot :**
- To really see the difference between a simulated environment and the real world

Thank you for your attention !

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